Lecture 31 Canonical Correlation I: Population

STAT 441: Statistical Methods for Learning and Data Mining

Wenlu TANG

Department of Mathematical and Statistical Sciences
University of Alberta

Winter, 2026

Outline

1. Purpose of Canonical Correlation Analysis
2. Population Canonical Correlations
3. Example
4. Summary and Remark

Purpose of Canonical Correlation Analysis

- Canonical Correlation Analysis (CCA) connects two sets of variables by finding linear combinations of variables that maximally correlate.
- There are two typical purposes of CCA:
  - Data reduction: explain covariation between two sets of variables using small number of linear combinations
  - Data interpretation: find features (i.e., canonical variates) that are important for explaining covariation between sets of variables

UNIVERSITY OF ALBERTA
EDMONTON·ALBERTA·CANADA

Continued

- Canonical Correlation Analysis (CCA) is a dimension reduction method that is similar to PCA, but where we simultaneously reduce the dimension of two random vectors $\mathbf{y}$ and $\mathbf{x}$
- Instead of trying to explain overall variance, we try to explain the covariance $\mathrm{Cov}(\mathbf{y}, \mathbf{x})$
  - Note that this is a measure of association between $\mathbf{y}$ and $\mathbf{x}$
- Example include:
  - Arithmetic speed and power ($\mathbf{y}$) and reading speed and power ($\mathbf{x}$)
  - College performance metrics ($\mathbf{y}$) and high-school achievement metrics ($\mathbf{x}$)

Population Canonical Correlations

- Given two random vectors $\mathbf{x}_{p\times 1}, \mathbf{y}_{q\times 1}$ ($p \leq q$ by convention) with covariance
$$\text{COV}\left[\begin{pmatrix} \mathbf{x} \\ \mathbf{y} \end{pmatrix}\right] = \boldsymbol{\Sigma} = \begin{pmatrix} \boldsymbol{\Sigma}_{11} & \boldsymbol{\Sigma}_{12} \\ \boldsymbol{\Sigma}_{21} & \boldsymbol{\Sigma}_{22} \end{pmatrix},$$
(with $rk\left(\boldsymbol{\Sigma}_{11}\right) = rk\left(\boldsymbol{\Sigma}_{12}\right) = p$, $rk\left(\boldsymbol{\Sigma}_{22}\right) = q$),
- we wish to summarize the relationships between the two sets of variables.

- Let $U = \boldsymbol{a}^T\mathbf{x}$ and $V = \boldsymbol{b}^T\mathbf{y}$ be linear combinations of $\mathbf{x}$ and $\mathbf{y}$, respectively.
- We have:
  - $\mathrm{Var}(U) = \boldsymbol{a}^T\boldsymbol{\Sigma}_{11}\boldsymbol{a}$
  - $\mathrm{Var}(V) = \boldsymbol{b}^T\boldsymbol{\Sigma}_{22}\boldsymbol{b}$
  - $\mathrm{Cov}(U, V) = \boldsymbol{a}^T\boldsymbol{\Sigma}_{12}\boldsymbol{b}$.
- First find $\mathbf{a}_1, \mathbf{b}_1$ to maximize

$$\mathrm{CORR}\left[\mathbf{a}_1^T\mathbf{x}, \mathbf{b}_1^T\mathbf{y}\right] = \frac{\mathbf{a}_1^T\boldsymbol{\Sigma}_{12}\mathbf{b}_1}{\sqrt{\mathbf{a}_1^T\boldsymbol{\Sigma}_{11}\mathbf{a}_1}\sqrt{\mathbf{b}_1^T\boldsymbol{\Sigma}_{22}\mathbf{b}_1}}.$$

- This is the *first pair of canonical variables*, or *first canonical variate pair*.

Population Canonical Correlations

- The second pair is $\left(\mathbf{a}_2^T\mathbf{x}, \mathbf{b}_2^T\mathbf{y}\right)$, chosen to maximize the correlation subject to the requirement that

$$\text{CORR}\left[\mathbf{a}_1^T\mathbf{x}, \mathbf{a}_2^T\mathbf{x}\right] = \text{CORR}\left[\mathbf{b}_1^T\mathbf{y}, \mathbf{b}_2^T\mathbf{y}\right] = 0,$$

  i.e.

$$\mathbf{a}_1^T\boldsymbol{\Sigma}_{11}\mathbf{a}_2 = \mathbf{b}_1^T\boldsymbol{\Sigma}_{22}\mathbf{b}_2 = 0.$$

- Continue, ... , p pairs.

Solutions

- First standardize: define

$$\begin{pmatrix} \tilde{\boldsymbol{x}} \\ \tilde{\boldsymbol{y}} \end{pmatrix} = \begin{pmatrix} \boldsymbol{\Sigma}_{11}^{-1/2}\mathbf{x} \\ \boldsymbol{\Sigma}_{22}^{-1/2}\mathbf{y} \end{pmatrix},$$

with

$$\text{cov}\left[\begin{pmatrix} \tilde{\boldsymbol{x}} \\ \tilde{\boldsymbol{y}} \end{pmatrix}\right] = \begin{pmatrix} \mathbf{I}_p & \tilde{\boldsymbol{\Sigma}}_{12} \\ \tilde{\boldsymbol{\Sigma}}_{21} & \mathbf{I}_q \end{pmatrix},$$

where

$$\tilde{\boldsymbol{\Sigma}}_{12} = \boldsymbol{\Sigma}_{11}^{-1/2}\boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1/2} = \tilde{\boldsymbol{\Sigma}}_{21}^{T}.$$

Solutions

- Let
$$\tilde{\mathbf{\Sigma}}_{12} = \mathbf{G}_{p\times p}\mathbf{R}_{p\times q}\mathbf{H}^T_{q\times q}$$
be the svd, with
$$\mathbf{R} = \begin{pmatrix} \rho_1^* & & 0 & 0 & \cdots & 0 \\ & \ddots & & \vdots & & \vdots \\ 0 & & \rho_p^* & 0 & \cdots & 0 \end{pmatrix},$$
and $\rho_1^* \geq \cdots \rho_p^* > 0$.
- Since
$$\mathbf{R}\mathbf{R}^T = \mathbf{G}^T\tilde{\mathbf{\Sigma}}_{12}\tilde{\mathbf{\Sigma}}_{21}\mathbf{G},$$
is diagonal, its diagonal elements ρ_i^{*2} are the eigenvalues, and the columns $\{\mathbf{g}_i\}$ of $\mathbf{G}$ are the corresponding orthonormal eigenvectors, of
$$\tilde{\mathbf{\Sigma}}_{12}\tilde{\mathbf{\Sigma}}_{21} = \mathbf{\Sigma}_{11}^{-1/2}\mathbf{\Sigma}_{12}\mathbf{\Sigma}_{22}^{-1}\mathbf{\Sigma}_{21}\mathbf{\Sigma}_{11}^{-1/2}.$$

Solutions

- The ρ_i^{*2} are also the nonzero eigenvalues of $\tilde{\Sigma}_{21}\tilde{\Sigma}_{12} : q \times q$, and the columns of $\mathbf{H}$ are the corresponding orthonormal eigenvectors.
- Now put

$$\begin{aligned} \tilde{\boldsymbol{u}}_{p\times 1} &= \mathbf{G}^T\tilde{\boldsymbol{x}} = \mathbf{G}^T\boldsymbol{\Sigma}_{11}^{-1/2}\mathbf{x}, \\ \tilde{\boldsymbol{v}}_{q\times 1} &= \mathbf{H}^T\tilde{\boldsymbol{y}} = \mathbf{H}^T\boldsymbol{\Sigma}_{22}^{-1/2}\mathbf{y}, \end{aligned}$$

(similar to pc's) with

$$\text{COV}\left[\begin{pmatrix} \tilde{\boldsymbol{u}} \\ \tilde{\boldsymbol{v}} \end{pmatrix}\right] = \begin{pmatrix} \mathbf{I}_p & \mathbf{R} \\ \mathbf{R}^T & \mathbf{I}_q \end{pmatrix}.$$

Solutions

- Thus, with $\{\mathbf{g}_i\}$ the columns of $\mathbf{G}$ and $\{\mathbf{h}_i\}$ the columns of $\mathbf{H}$, the canonical pairs (maximizing the correlations, and uncorrelated with each other) are, for $i = 1, ..., p$,

$$\left\{\left(\tilde{U}_i, \tilde{V}_i\right)\right\} = \left\{\left(\mathbf{g}_i^T \boldsymbol{\Sigma}_{11}^{-1/2}\mathbf{x}, \mathbf{h}_i^T \boldsymbol{\Sigma}_{22}^{-1/2}\mathbf{y}\right)\right\}. \tag{2.1}$$

- The ith canonical correlation is

$$\text{CORR}\left[\tilde{U}_i, \tilde{V}_i\right] = \rho_i^*,$$

(the singular values of $\tilde{\boldsymbol{\Sigma}}_{12}$), and

$$\text{VAR}\left[\tilde{U}_i\right] = \text{VAR}\left[\tilde{V}_i\right] = 1.$$

Solutions

- It is often simpler to carry out the discussion in terms of the standardized variables $\tilde{\boldsymbol{x}}$ and $\tilde{\boldsymbol{y}}$, for which the canonical pairs are $\left\{\left(\mathbf{g}_i^T\tilde{\boldsymbol{x}}, \mathbf{h}_i^T\tilde{\boldsymbol{y}}\right)\right\}_{i=1}^p$ and the canonical correlations are the ρ_i^*.
- Note that $\|\mathbf{g}_i\| = \|\mathbf{h}_i\| = 1$, $i = 1, \ldots, p$. We sometimes work instead with the pairs $\left(\mathbf{a}_i^T\mathbf{x}, \mathbf{b}_i^T\mathbf{y}\right)$ and assume that $\|\mathbf{a}_i\| = \|\mathbf{b}_i\| = 1$, $i = 1, \ldots, p$. To enforce this, we normalize:

$$U_i = \mathbf{a}_i^T\mathbf{x},\ V_i = \mathbf{b}_i^T\mathbf{y};$$

- here (from (2.1))

$$\begin{aligned}
\mathbf{a}_i &= \boldsymbol{\Sigma}_{11}^{-1/2}\mathbf{g}_i \Big/ \left\|\boldsymbol{\Sigma}_{11}^{-1/2}\mathbf{g}_i\right\|, \\
\mathbf{b}_i &= \boldsymbol{\Sigma}_{22}^{-1/2}\mathbf{h}_i \Big/ \left\|\boldsymbol{\Sigma}_{22}^{-1/2}\mathbf{h}_i\right\|.
\end{aligned}$$

Solutions

- This does not affect the correlations but

$$\begin{aligned} \text{VAR}\left[U_i\right] &= 1/\left\|\boldsymbol{\Sigma}_{11}^{-1/2}\mathbf{g}_i\right\|^2 = 1/\left[\mathbf{G}^T\boldsymbol{\Sigma}_{11}^{-1}\mathbf{G}\right]_{ii}, \\ \text{VAR}\left[V_i\right] &= 1/\left\|\boldsymbol{\Sigma}_{22}^{-1/2}\mathbf{h}_i\right\|^2 = 1/\left[\mathbf{H}^T\boldsymbol{\Sigma}_{22}^{-1}\mathbf{H}\right]_{ii}. \end{aligned}$$

- Note that

$$\begin{aligned} \sum_{i=1}^{p}\text{VAR}\left[X_i\right] &= tr\boldsymbol{\Sigma}_{11} = \sum_{i=1}^{p}\left[\mathbf{G}^T\boldsymbol{\Sigma}_{11}\mathbf{G}\right]_{ii} \\ &\overset{why?}{\geq} \sum_{i=1}^{p}\text{VAR}\left[U_i\right], \end{aligned} \tag{2.2}$$

hence there is no decomposition of the total variance (of $\mathbf{x}$) here.

Example

- There are examples in which the first few canonical variables account for very little variability in each set; e.g.

$$\boldsymbol{\Sigma} = \begin{pmatrix} 100 & 0 & 0 & 0 \\ 0 & 1 & .95 & 0 \\ 0 & .95 & 1 & 0 \\ 0 & 0 & 0 & 100 \end{pmatrix};$$

- Here the highly correlated variables have very little variation: the first canonical pair is (X_2, Y_1) and the second is (X_1, Y_2).
- Hence we do have equality in (2.2) but $\text{VAR}[U_1] = \text{VAR}[V_1] = 1$ only.

Calculation

- We have $p = q = 2$;

$$\begin{aligned}\tilde{\boldsymbol{\Sigma}}_{12} &= \boldsymbol{\Sigma}_{11}^{-1/2}\boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1/2} \\ &= \begin{pmatrix} 1/10 & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 0 & 0 \\ .95 & 0 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & 1/10 \end{pmatrix} \\ &= \begin{pmatrix} 0 & 0 \\ .95 & 0 \end{pmatrix};\end{aligned}$$

- and

$$\begin{aligned}\tilde{\boldsymbol{\Sigma}}_{12}\tilde{\boldsymbol{\Sigma}}_{21} &= \begin{pmatrix} 0 & 0 \\ 0 & (.95)^2 \end{pmatrix}; \\ \rho_1^* &= .95,\ \rho_2^* = 0; \\ \mathbf{g}_1 &= \begin{pmatrix} 0 \\ 1 \end{pmatrix},\ \mathbf{g}_2 = \begin{pmatrix} 1 \\ 0 \end{pmatrix};\end{aligned}$$

Calculation

- and

$$
\begin{aligned}
\tilde{\boldsymbol{\Sigma}}_{21}\tilde{\boldsymbol{\Sigma}}_{12} &= \begin{pmatrix} (.95)^2 & 0 \\ 0 & 0 \end{pmatrix}; \\
\mathbf{h}_1 &= \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \ \mathbf{h}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}; \\
\left(\tilde{U}_1, \tilde{V}_1\right) &= \left(\tilde{X}_2, \tilde{Y}_1\right) \propto \left(X_2, Y_1\right) = \left(U_1, V_1\right); \\
\left(\tilde{U}_2, \tilde{V}_2\right) &= \left(\tilde{X}_1, \tilde{Y}_2\right) \propto \left(X_1, Y_2\right) = \left(U_2, V_2\right).
\end{aligned}
$$

Summary and Remark

- Population Canonical Correlation
- Example